Pradeep Kumar Sharma

Senior Developer & Architect @ IBM ISDL
Storage Systems, Data Protection, Backup & Recovery

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https://github.com/Pradeep-Kumar-Sharma/profile

Profile Summary

19 Years experience | Storage Systems | Data Protection | Backup & Recovery | FinTech World | Trading Systems | HFT | Capital Market| Digital Signage | Game Development | Enterprise Software Products | AI-Assisted Engineering

Roles Performed: Architect, Engineering Manager, Product Owner, Senior Technical Manager, Project Manager, Scrum Master, L2/L3 Production Support & Sustenance Lead/Engineer, Individual Contributor (IC)

Senior software engineering professional with 19 years of experience in building and delivering mission-critical enterprise software products across Enterprise Storage and Data Protection, Backup & Recovery, FinTech world, High-Frequency Trading (HFT), Algorithmic Trading, Cybersecurity, Infrastructure Management, Device Management, Distributed Systems, Virtualization, and Cloud Platforms. Experienced across individual contributor, technical leadership, product ownership, project management, and support engineering roles, collaborating effectively with global engineering, product, business, marketing, and operations stakeholders to deliver innovative, scalable, and customer-focused solutions across diverse geographies and mission-critical environments. Leading cross-functional engineering teams, managing critical customer and production environments, and delivering business outcomes for global organizations including IBM, Goldman Sachs, Hold Brothers, Microsoft, Symantec, McAfee, Lenovo, Sennheiser, and NetIQ.

AI & AI-Assisted Engineering: Proficient in AI-assisted software engineering practices, leveraging generative AI technologies and modern developer productivity tools including IBM Bob, IBM watsonX, Claude Code, GitHub Copilot, OpenAI Codex, and Cursor to accelerate Software Architecture, Design, Development, Code Generation, Troubleshooting, Root-Cause Analysis, Test Automation, Technical Documentation, Project Planning, Deployment, Production Support. Experienced in building and integrating Model Context Protocol (MCP) Servers, Developing AI Agents & Tools, Automate Workflows, Developer Productivity Tools.

Security Engineering & Product Security: Application and Product Security, Secure SDLC, Secure Coding Practices, Threat Modeling, Vulnerability Management, Static Application Security Testing (SAST), Dynamic Application Security Testing (DAST), Software Composition Analysis (SCA), Cryptography, PKI, TLS/SSL, X.509 Certificates, Certificate Lifecycle Management, Post-Quantum Cryptography (PQC), Quantum-Safe Cryptography, Security Hardening

Core Domains & Technical Specialisation

Domain-1: Storage Systems | Data Protection | Backup & Recovery | (My current assignments @ IBM)

Designing, developing, and sustaining mission-critical enterprise backup and recovery platforms, data protection solutions, disaster recovery systems, enterprise storage, cloud ecosystems, and virtualization technologies. Focused on architecting and enhancing IBM Storage Protect (SP) and IBM Storage Protect Plus (SPP) and their and long-term product sustenance.

Domain Keywords: Enterprise Backup & Recovery · Data Protection & Data Resilience · Disaster Recovery (DR) · Storage Management · Enterprise Storage Platforms · VMware vSphere · vCenter · ESXi · VMware VDDK/VADP · VMware Cloud Foundation (VCF) · Hyper-V & VSS · Change Block Tracking (CBT) · Snapshot Management · Virtual Machine Protection · SaaS Backup · Agent-Based & Agentless Backup · Object Storage & S3 Storage · Cloud Storage & Cloud Tiering · Data Deduplication · Long-Term Retention · Air-Gapped & Immutable Backups · Ransomware Resilience · Backup Infrastructure · Storage Security · Data Lifecycle Management · RPO/RTO Optimization · Production Support & Root Cause Analysis · Linux & Windows Platforms · Distributed Systems · resolving complex production issues.

Companies: Architect at **IBM ISDL** (India Systems Development Lab) | Senior Developer at **Seagate** Toolkit

Key Contributions:

1. Architecture, development, and sustenance of IBM Storage Protect (SP) and IBM Storage Protect Plus (SPP).
2. Design and enhancement of backup, restore, recovery, and disaster recovery workflows for enterprise customers.
3. VMware integration leveraging vSphere, vCenter, ESXi, VDDK/VADP, VSS, and VMware Cloud Foundation (VCF).
4. Design and troubleshooting of virtual-machine backup, snapshot, and restore solutions.
5. Development of scalable storage and data management capabilities supporting large-scale enterprise deployments.
6. Resolution of complex production issues through debugging, root-cause analysis, customer-environment investigation, and cross-functional collaboration.
7. Delivery of security-focused features including certificate lifecycle management and storage security enhancements.
8. Collaboration with product management, customer support, sustainment, and engineering teams to drive product quality and customer success.

Domain-2: FinTech World | High-Frequency Trading (HFT) | Algorithmic Trading | Low-Latency Trading | Electronic Communication Networks (ECNs) | FIX Protocol | Equity/ Capital Markets

Designing, developing, and supporting mission-critical software systems across various enterprise FinTech Platforms and Software Products. Experience building and sustaining trading platforms supporting trade across global markets. Strong background trader-facing application development, risk-sensitive production support, and business continuity planning (BCP). Proven ability to partner closely with traders, quantitative teams, product managers, and business stakeholders to deliver scalable, resilient, and latency-sensitive trading solutions. Experience in end-to-end ownership from architecture and development through deployment, performance tuning, and production support in highly regulated and performance-critical financial environments.

Domain Keywords: Capital Markets & Trading Technologies: Capital Markets & Trading Expertise · Algorithmic Trading · High-Frequency Trading (HFT) . Electronic Trading · Equity & FX · Capital Markets Infrastructure · Low-Latency Systems · Real-Time Order Routing · Trading Automation . Order Management Systems (OMS) · Execution Management Systems (EMS) . Market Connectivity · Exchange Connectivity · ECN Connectivity & Connectivity · Order Lifecycle Management . Financial Messaging . FIX Protocol (4.2 / 4.4) . Market Data Processing . Trade Execution . Risk Controls & Validation . Trader Workflows · Trade Lifecycle Management · Production Trading Support . Business Continuity Planning (BCP) / Mock Trading & Disaster Recovery . Performance Tuning & Latency Profiling · Multithreading & Synchronization · Concurrent Programming . Lock-Free Architectures · Lock-Free Data Structures · Latency Profiling & Optimization · Memory Optimization . Linux & Windows Platforms . Distributed Systems· SecDB/Slang (Goldman Sachs proprietary)

Companies: Vice President, Equity Technology at **Goldman Sachs** | Senior Developer at **Hold Brothers Capital** | **NOMURA**

Key Contributions:

1. Developed and enhanced the DART algorithmic trading platform supporting equity traders across multiple Asia-Pacific markets.
2. Partnered directly with front-office traders and business stakeholders to rapidly deliver trading functionality and workflow enhancements.
3. Designed and maintained latency-sensitive components supporting order generation, routing, execution, and monitoring.
4. Delivered end-to-end ownership across architecture, development, testing, deployment, and production support.
5. Led critical production incident resolution and operational support for real-time trading environments.
6. Participated in Business Continuity Planning (BCP) and mock trading exercises to ensure operational resilience.
7. Collaborated with global technology teams to improve platform stability, scalability, and performance.

Domain-3: Software Application Development (Device, Asset, Security, Signage & Gaming)

Product Engineering, Application & System Architect with experience across the Full Product Lifecycle, Architecture, Designing, Development, Production Debugging, and Long-Term Sustenance of Enterprise Software Products. Experience in Digital Signage, Game Development. Known for technical leadership, complex production issue resolution, and effective cross-functional collaboration across engineering, product, and operations teams to deliver secure, reliable, maintainable, resilient, high-performance and high-quality software solutions for mission-critical enterprise software solutions.

Domain Keywords: Device Management · Asset Management · Endpoint Management · Device Lifecycle Management · Firmware Management · Infrastructure Monitoring · Remote Device Management · Enterprise Monitoring · Data-Center Monitoring · IT Operations · Systems Observability · IT Automation · Endpoint Security · Security Platforms · Cloud Applications · Client-Server Architecture · Production Support · Root-Cause Analysis · Win32 · MFC · COM/ATL · Microsoft DirectX · STL · ATL · C#.NET · WMI · CIM · WBEM · WinRM · PowerShell · Cross-Platform Development (Windows, macOS, Linux)

Companies: **Microsoft** | **Sennheiser/Epos** | **McAfee** | **Symantec** | **NetIQ** | **Lenovo** Cloud | **NAVORI** Signage | **Octavian** Gaming

Key Contributions:

1. *Architecture & Design:* Architected scalable enterprise software using HLD, LLD, OOAD, and UML.
2. *Cross-Platform Development:* Developed and sustained applications across Windows, macOS, and Linux using C/C++, Modern C++, Windows Programming, Win32 API, COM/ATL, MFC, .NET, and Python.
3. *Device & Asset Management:* Device lifecycle management, endpoint administration, and asset management solutions.
4. *Enterprise Security:* Developed and maintained endpoint security and enterprise security software.
5. *Infrastructure Monitoring:* Built monitoring and remote administration capabilities using WMI, CIM, WBEM, and WinRM.
6. *Cloud-Connected Solutions:* Cloud-based device monitoring, firmware management, and enterprise applications.

7. *Digital Signage & Gaming:* Developed and sustained software for digital signage and gaming platforms.
8. *Performance & Concurrency:* Applied multithreading, performance optimization, and systems programming to improve application efficiency.
9. *Production Debugging:* Resolved complex production issues through systematic troubleshooting and root-cause analysis.
10. *Automation & Quality:* Leveraged scripting and test automation to improve engineering efficiency and software quality.
11. *End-to-End Product Engineering:* Managed activities across requirements, design, development, testing, release, and product sustenance.
12. *Technical Leadership:* Guided engineering teams through design reviews, technical problem-solving, and cross-functional collaboration.

Key Technical Skills | Core Competencies | Engineering Tools

1. Programming Languages: C/C++, Modern C++ (11/14/17/20), Python, C#/.NET, Go (Golang), PowerShell, Shell Scripting, Java
2. *Systems Programming:* Win32 API, STL, MFC, COM/ATL, DirectX, Multithreading, Concurrency, Memory Management, Data Structures & Algorithms, IPC, Performance Optimization, Cross-Platform Development (Windows, Linux, macOS)
3. Architecture & Design: System Architecture, Application Architecture, Distributed Systems, OOAD, Design Patterns, HLD/LLD, Scalability, Reliability Engineering, Code Reviews, Refactoring, Test Automation.
4. Debugging & Performance Engineering: WinDbg, GDB, Production Debugging, Root Cause Analysis, Crash Analysis, Performance Profiling, Performance Tuning, Memory Leak Analysis, Customer Escalation Management, Product Sustenance.
5. Static Code Analysis: PREfast, FxCop, CppCheck, Coverity, SonarQube, Mend (WhiteSource), Detect-Secrets.
6. Dynamic Code Analysis: Visual Studio Diagnostic Tools, DevPartner BoundsChecker, DeepCheck, Parasoft Insure++, BullseyeCoverage.
7. Monitoring, Diagnostics & Management: PerfMon, WMI, WMIC, WQL, CIM, WBEM, WBEMTest, WinRM, WSMan, PowerShell Remoting, Sysinternals Suite (Process Explorer, Process Monitor), Performance Counters, Event Viewer, SNMP, Syslog, Linux system monitoring tools(top, vmstat, iostat, sar, ps, systemd)
8. DevOps & SecOps: CI/CD & CI/CT, Jenkins, Git, GitHub, JIRA, Confluence, SVN, Perforce, Maven, Gradle, CppUnit, VersionOne, CSF (Goldman Sachs).
9. Virtualization, Containers & Cloud Platforms: VMware Ecosystem (Workstation, Player, ThinApp, vSphere, vCenter, ESXi 7/8/9, VDDK, SDKs/APIs, VMware Cloud Foundation, VCF), Microsoft Hyper-V, Volume Shadow Copy Service (VSS), Docker, Kubernetes, OpenShift, Container Orchestration, Virtualization Platforms, Cloud-Native Infrastructure.
10. AI-Assisted Engineering: GitHub Copilot, Claude Code, Cursor, OpenAI Codex, IBM watsonx, IBM Bob, AI Agent Development, MCP Server Integration, Intelligent Automation.

Courses and Certifications

1. Programming with Google Go / Golang
2. Principles of Secure Coding
3. Programming for Everybody (Getting Started with Python)
4. Retrieving, Processing, and Visualizing Data with Python
5. Python Data Structures
6. Using Databases with Python
7. Using Python to Access Web Data
8. Blockchain Basics
9. Introduction to Blockchain Technologies
10. Smart Contracts
11. Getting Started with Go
12. Introduction to the MongoDB
13. BackEnd Performance Optimization
14. Intro to Large Language Models (LLMs)
15. Introduction to SAN and NAS Storage
16. Container Management With Podman Made Easy and Hands-On
17. Beginner Introduction to Containers, Docker, and Kubernetes
18. Golang for Beginners
19. Intro to MCP (Model Context Protocol)

Work Experience

#	Company	Designation	Duration
1.	IBM ISDL (India Systems Development Lab)	Senior Developer & Architect	Since July 2023 **(Current)**
2.	LTIMindtree (Formerly Mindtree)	Senior Technical Manager (Architect)	Aug 2018 to June 2023
3.	Goldman Sachs	Vice President (Equity Tech)	Jun 2015 to June 2018
4.	Micro Focus (Formerly Novell)	Specialist	Feb 2012 to Feb 2015
5.	Xchanging	Technical Lead (Project)	Jan 2011 to Feb 2012
6.	Wipro Technologies (for Symantec)	Technical Lead	Sep 2006 to Sep 2010

Organisation, Clients and Technology Companies worked with

IBM ISDL-Storage | Seagate | Sennheiser (EPOS) | Microsoft | McAfee | Symantec Norton | Goldman Sachs | NOMURA | HOLD Brothers | ITG Investment Research | PayPal | NetIQ (Micro Focus) | Lenovo Cloud | NAVORI | Octavian Gaming Solutions

Education and Academics

- MBA (Information Technology) from Guru Jambheshwar University of Science & Technology in 2006
- MCA (Master of Computer Applications) from DOEACC Centre (Formerly Regional Computer Centre), Chandigarh in 2001
- Diploma in Information & System Management from APTECH computers, Ambala Cantt. Haryana in 1998
- B.Sc. in Physics Honors from DHSK College, Dibrugarh University, Assam in 1996

Project Portfolio: For detail information on projects, technologies, responsibilities and contributions please visit my Project Lists

Thanks & Regards,
Pradeep Kumar Sharma